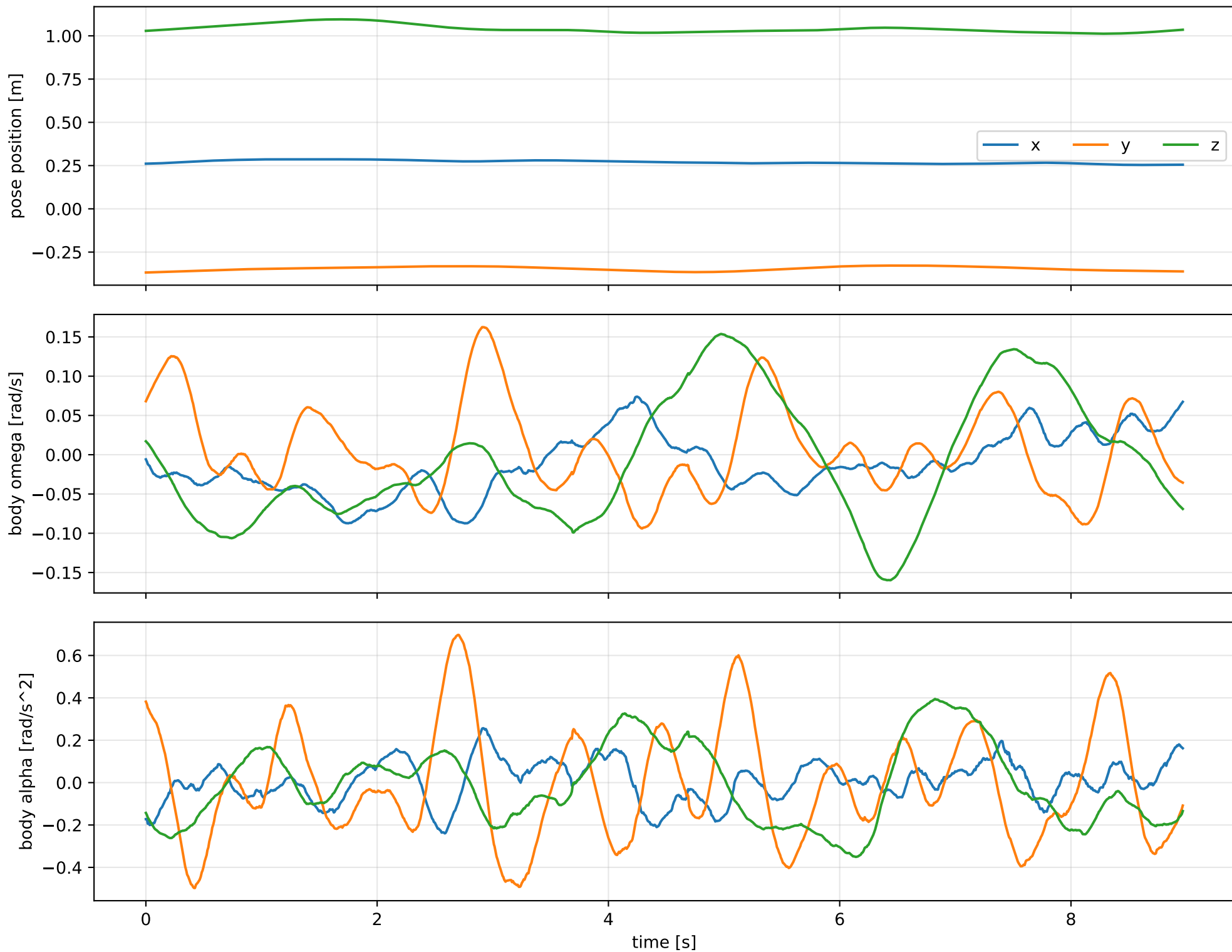
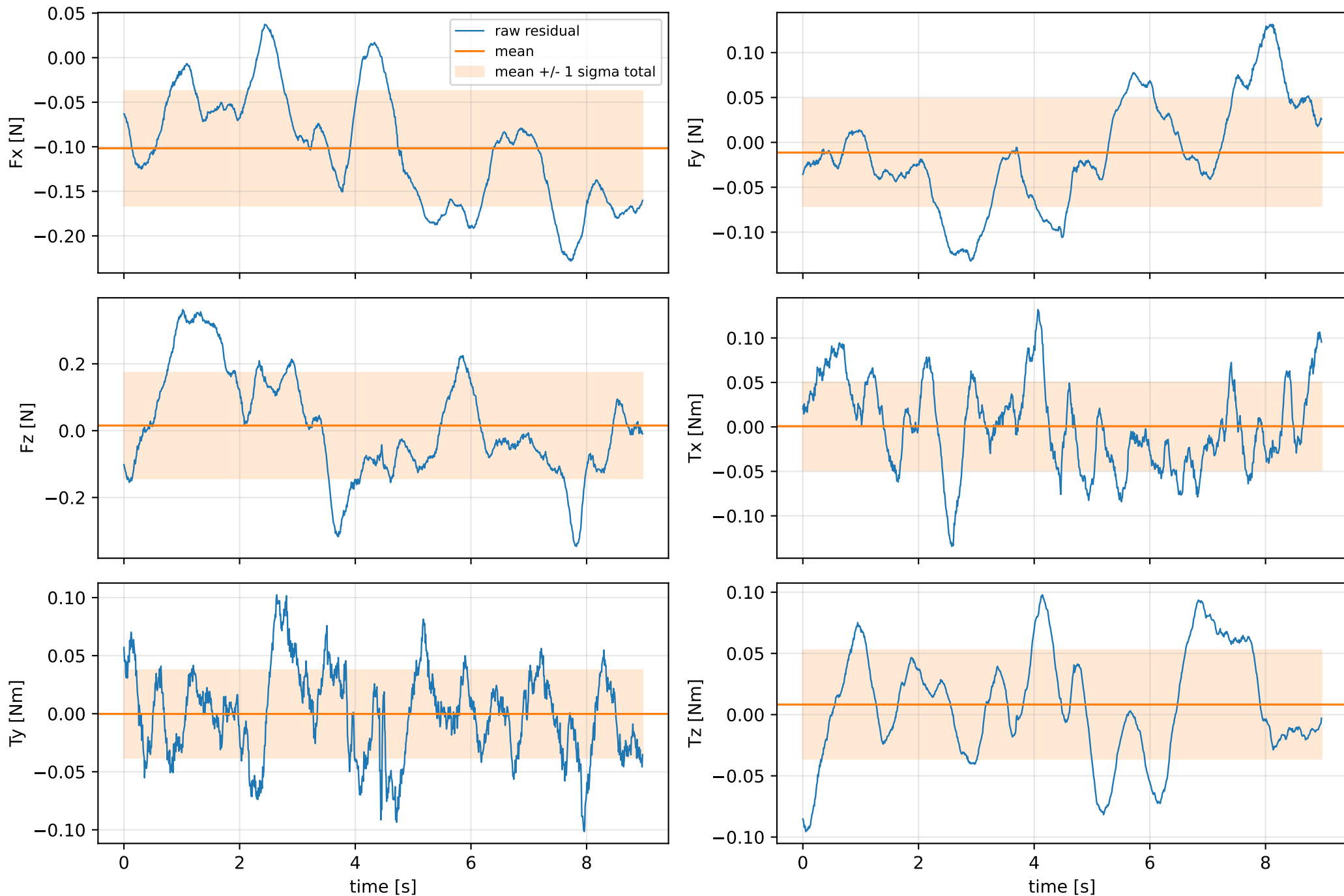


force_over_mass_rotor_1_nominal: SG trajectory and derived motion



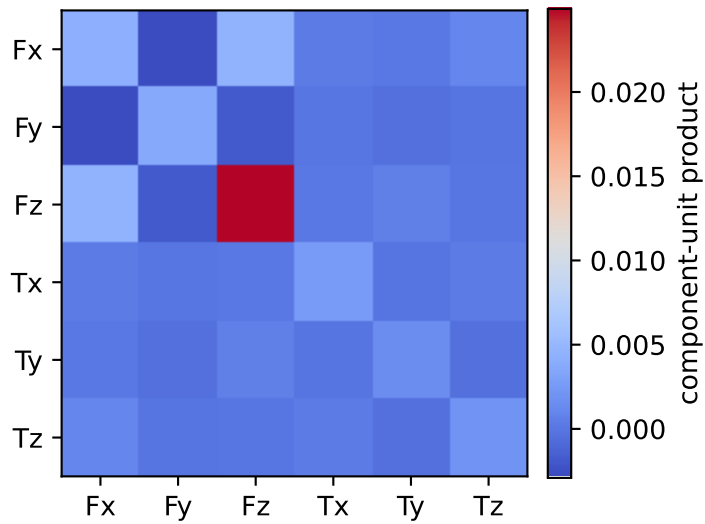
force_over_mass_rotor_1_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



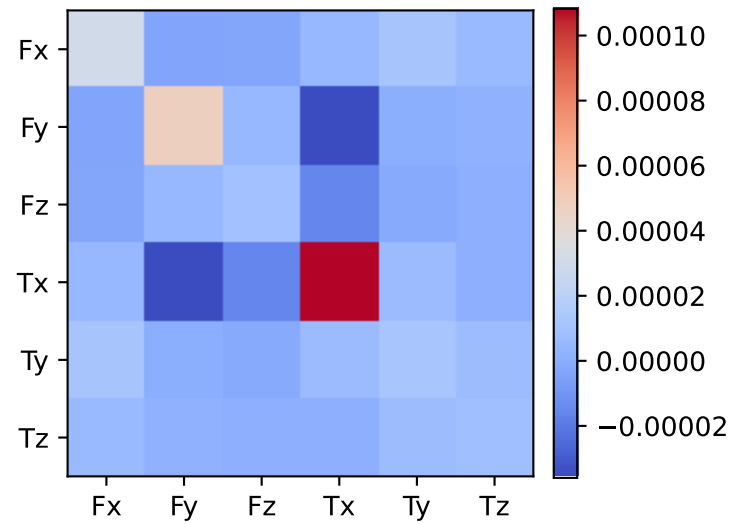
vector RMS about zero: force 0.2081 N, torque 0.076716 Nm; component std about mean: force [0.0644 0.0596 0.1579], torque [0.0493 0.0377 0.0444]

force_over_mass_rotor_1_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.002 0.003 0.006 0.026]

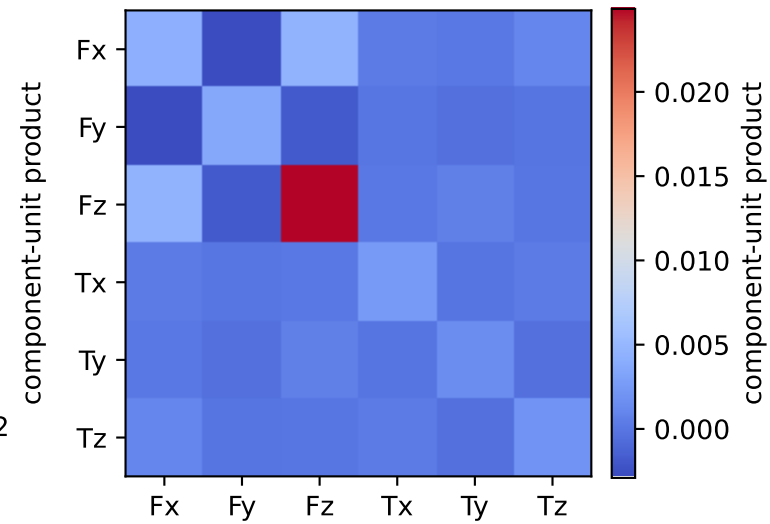
empirical total covariance



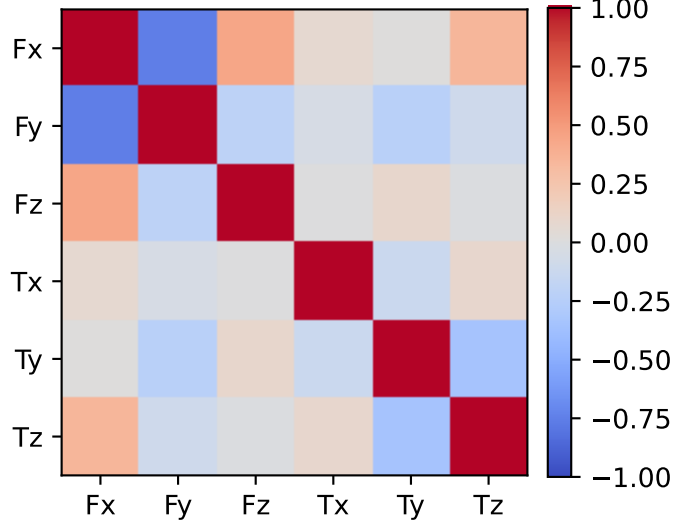
mean full-SG prediction covariance



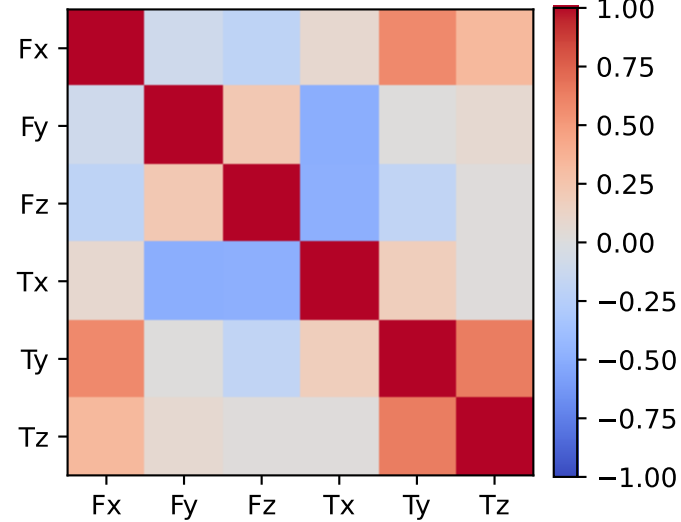
PSD excess model discrepancy covariance



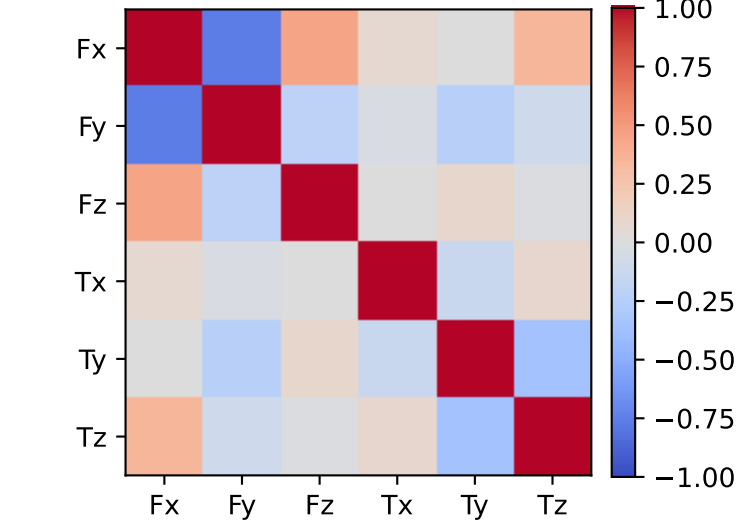
empirical total correlation



mean full-SG prediction correlation

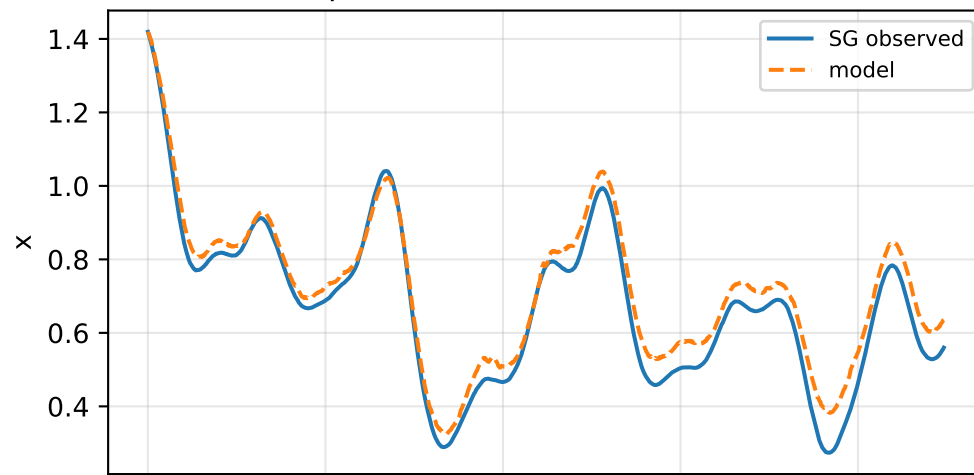


PSD excess model discrepancy correlation

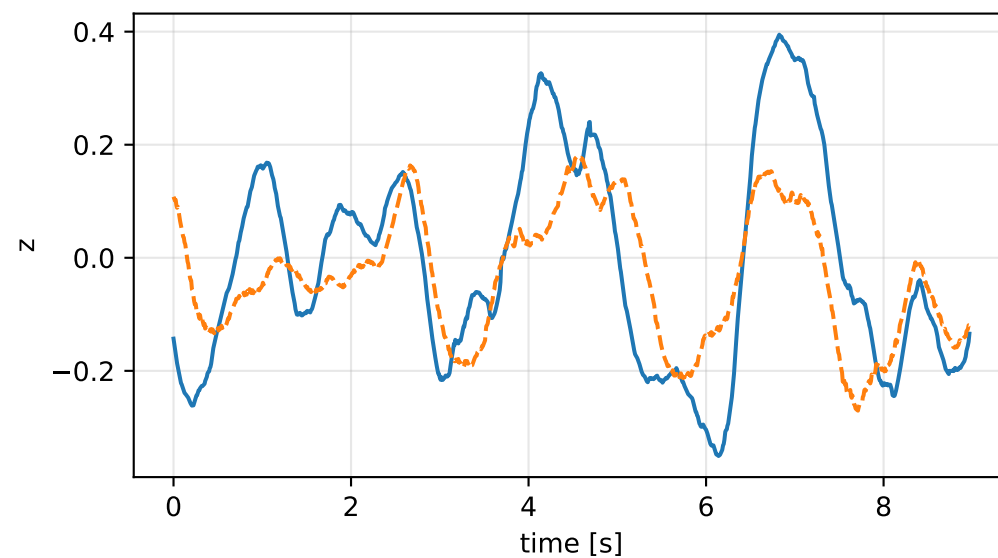
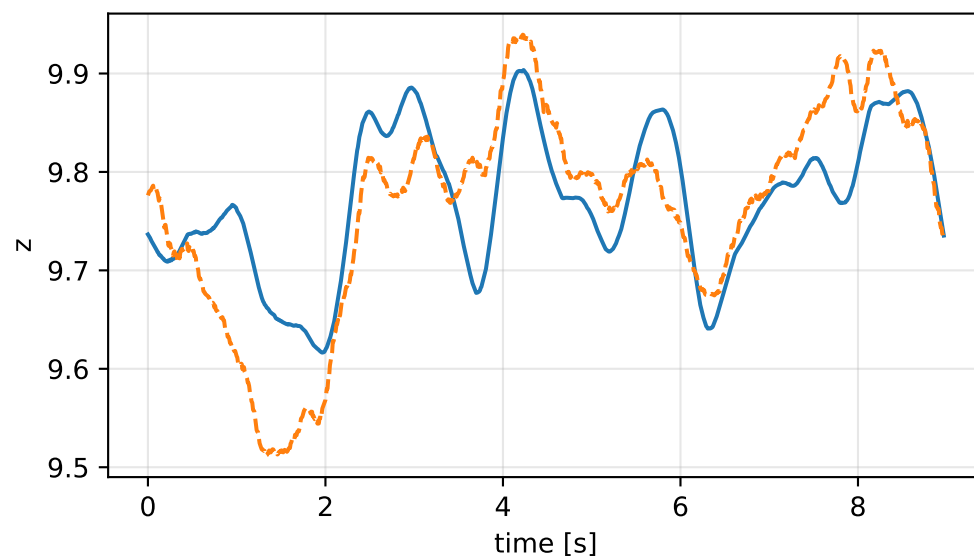
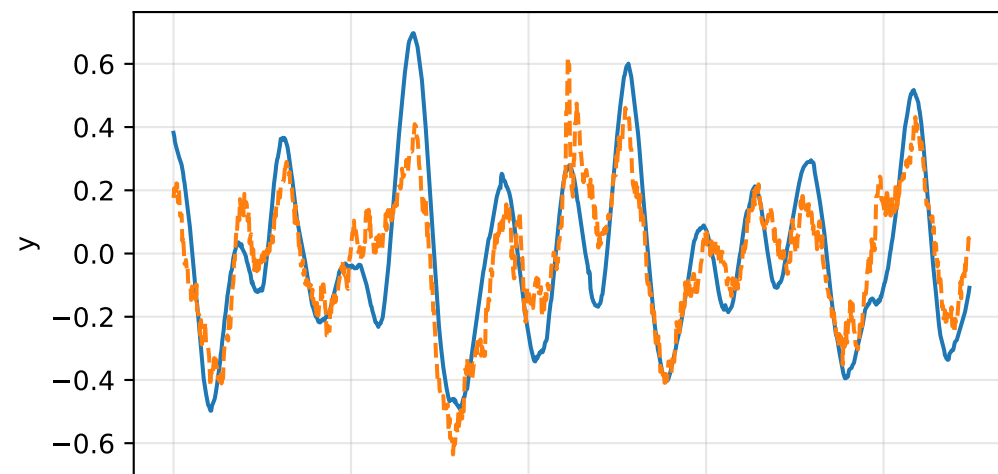
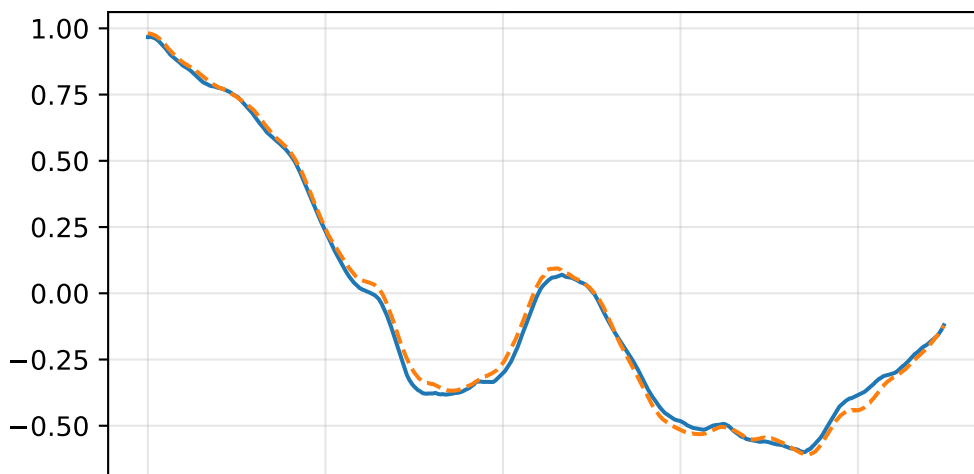
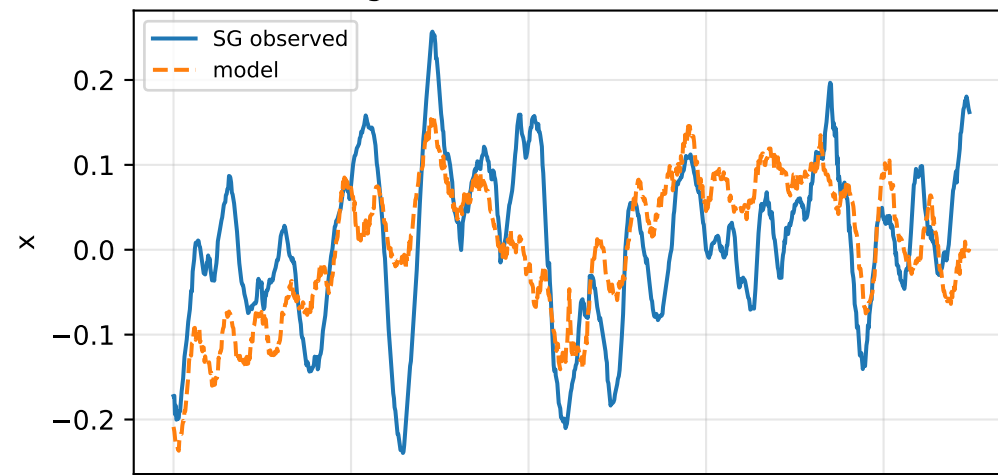


force_over_mass_rotor_1_nominal: acceleration residual objective

specific acceleration [m/s^2]

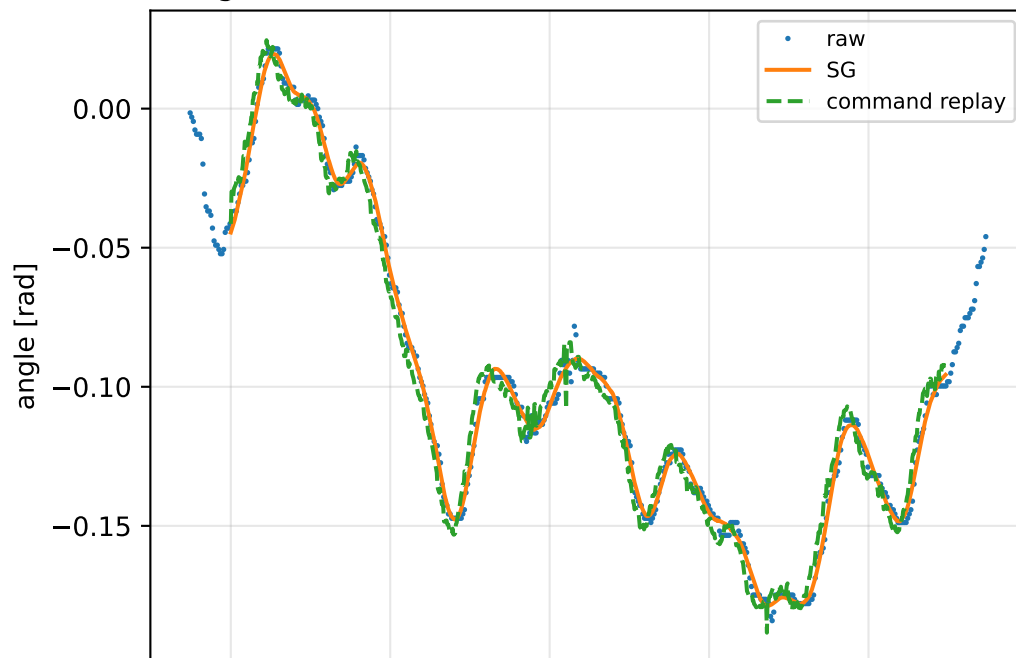


angular acceleration [rad/s^2]

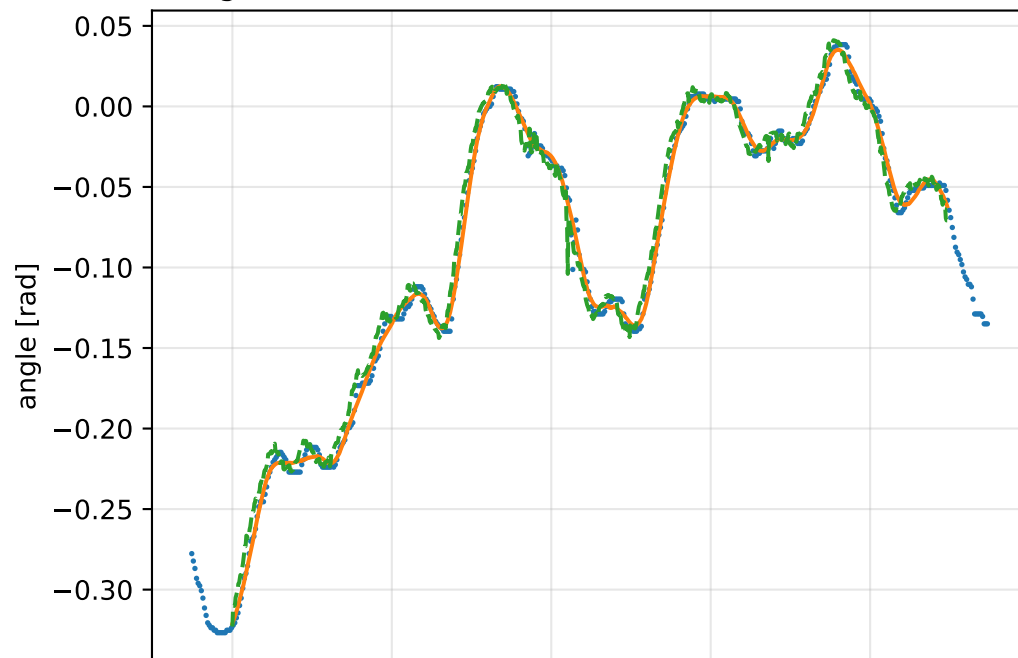


force_over_mass_rotor_1_nominal: gimbal measurement audit (objective=measured_sg)

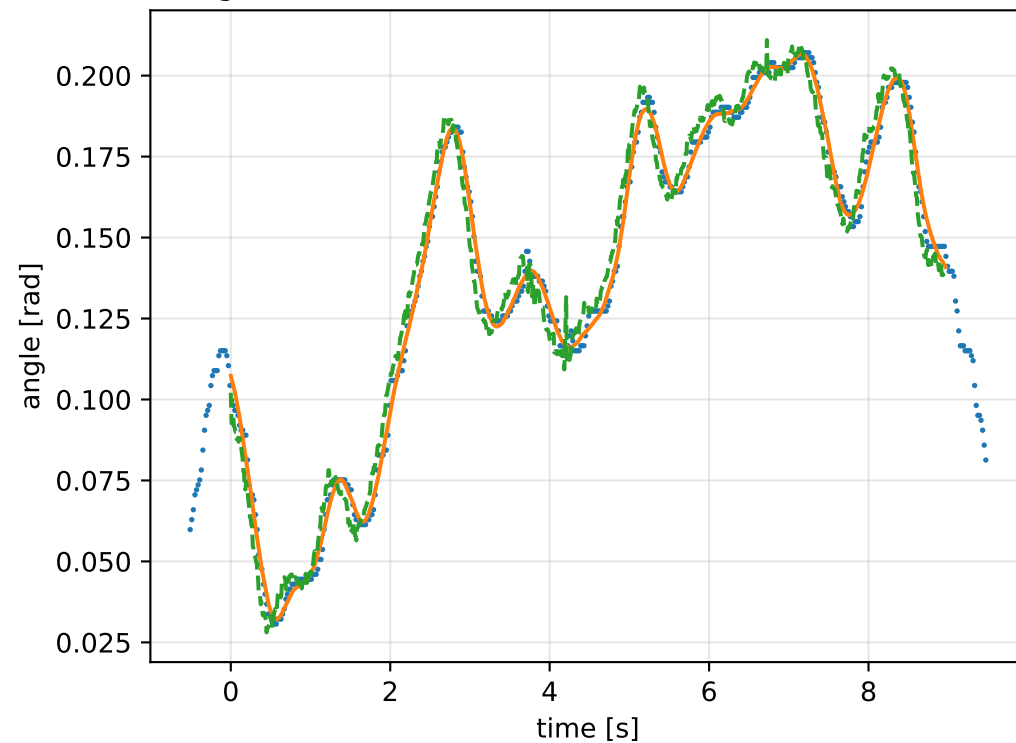
gimbal 1: RMSE=0.006345, max=0.01494 rad



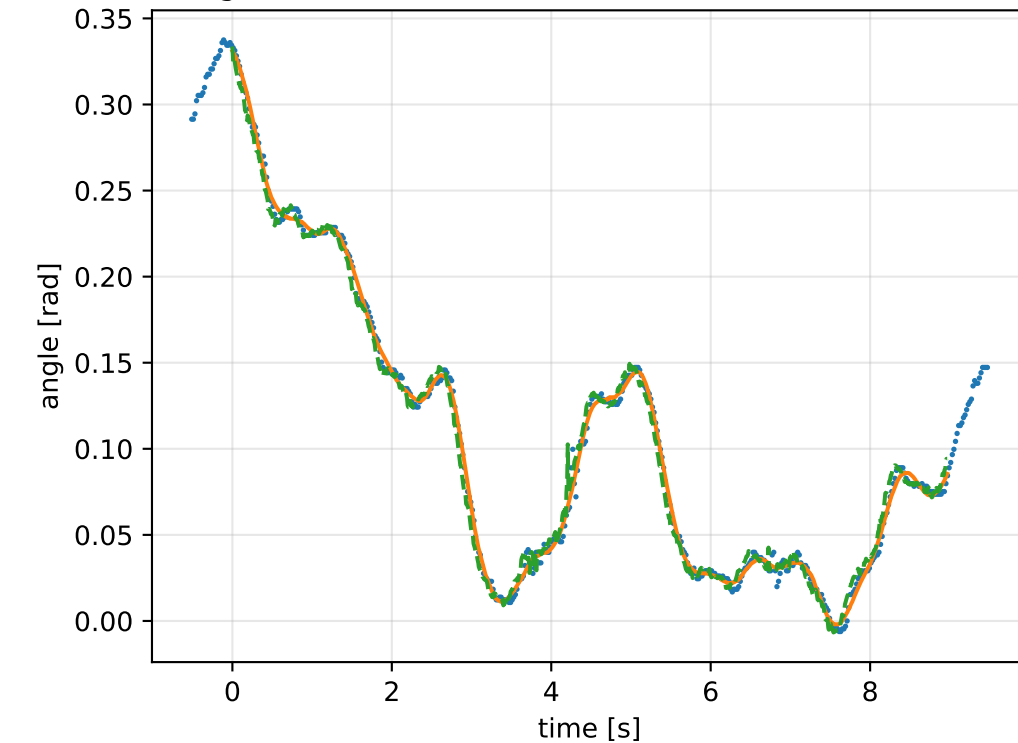
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

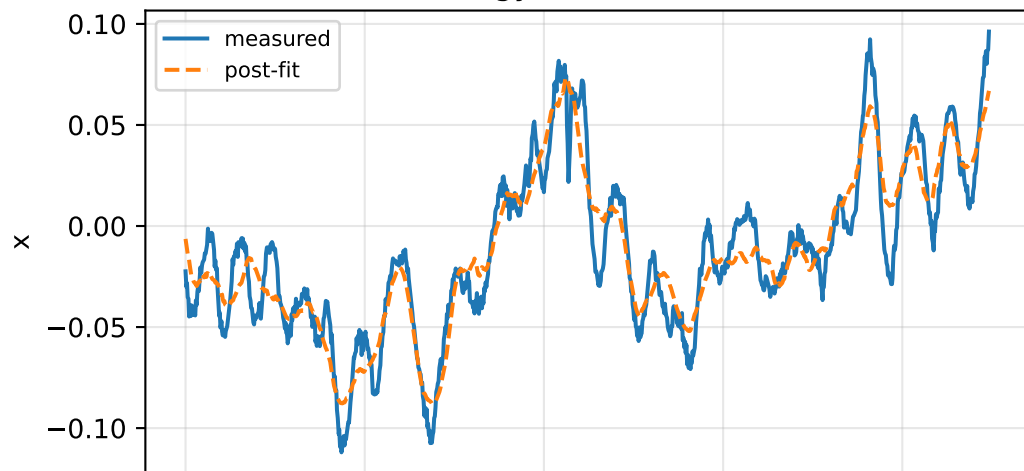


gimbal 4: RMSE=0.006321, max=0.03642 rad

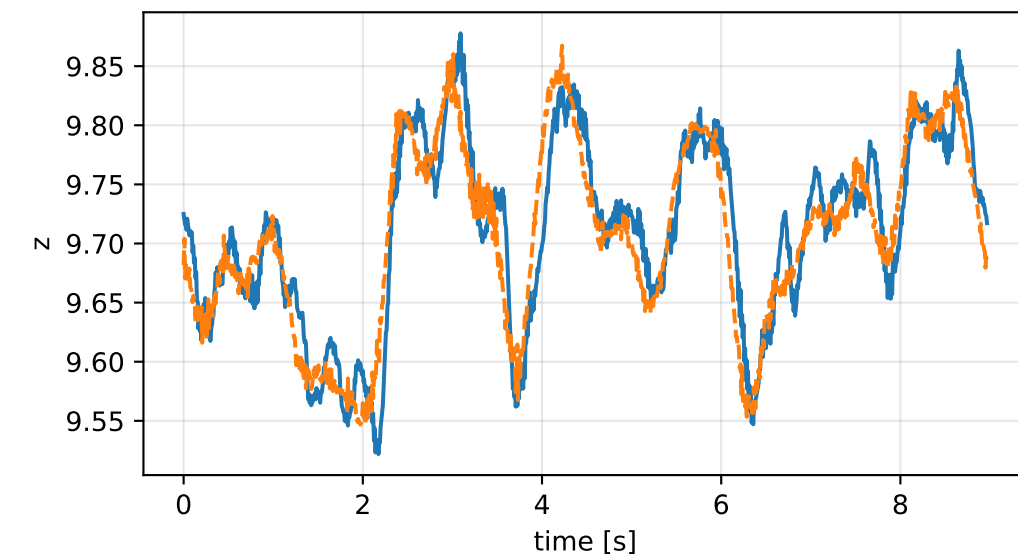
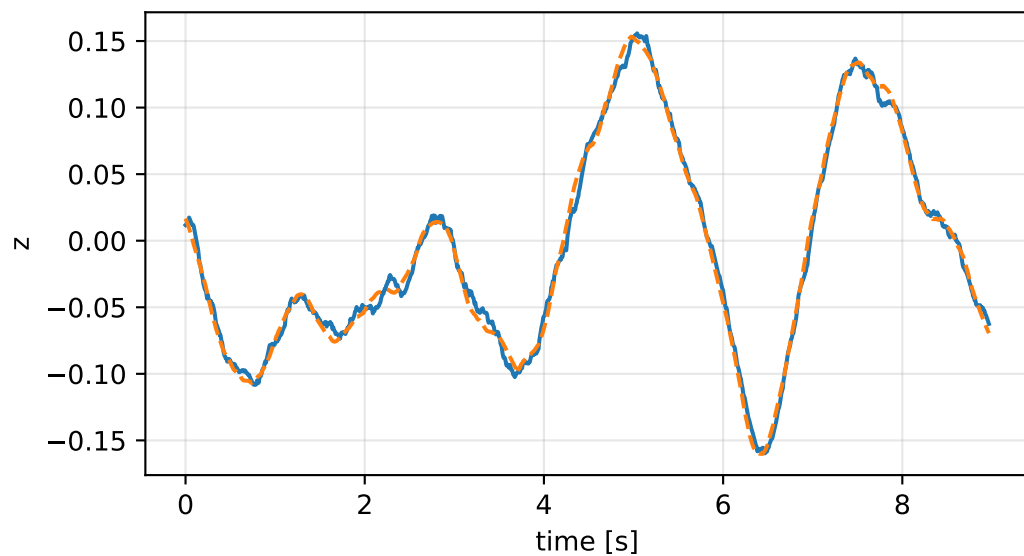
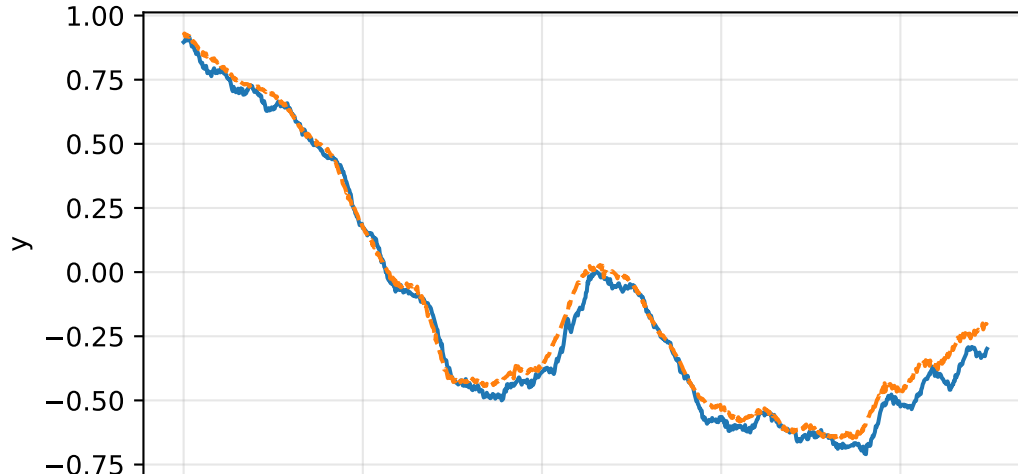
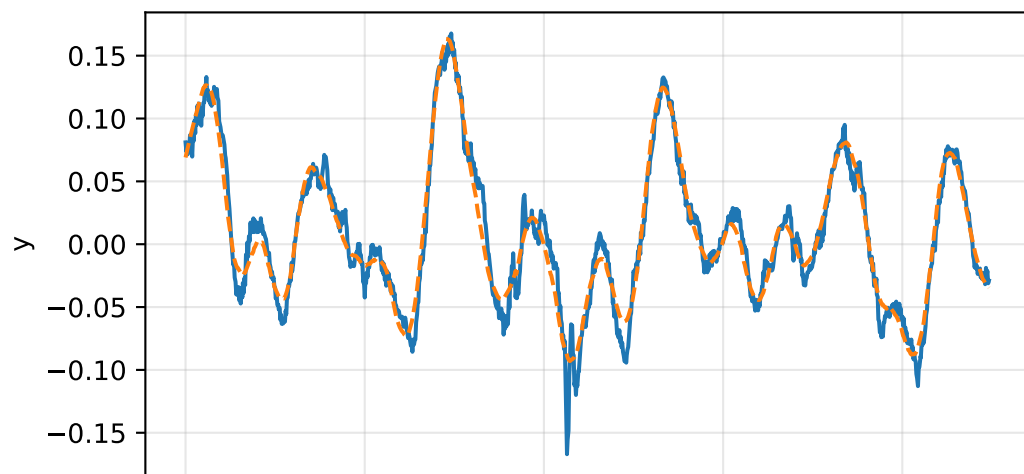
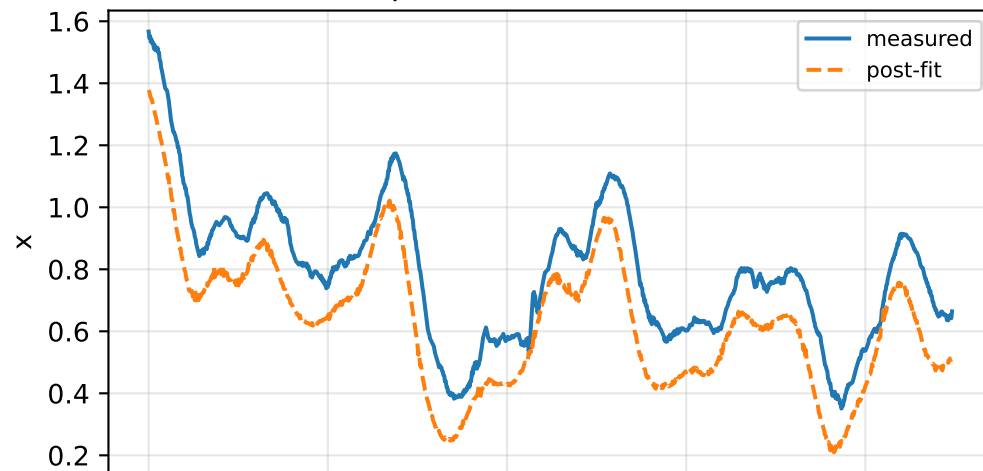


force_over_mass_rotor_1_nominal: IMU comparison (diagnostic only)

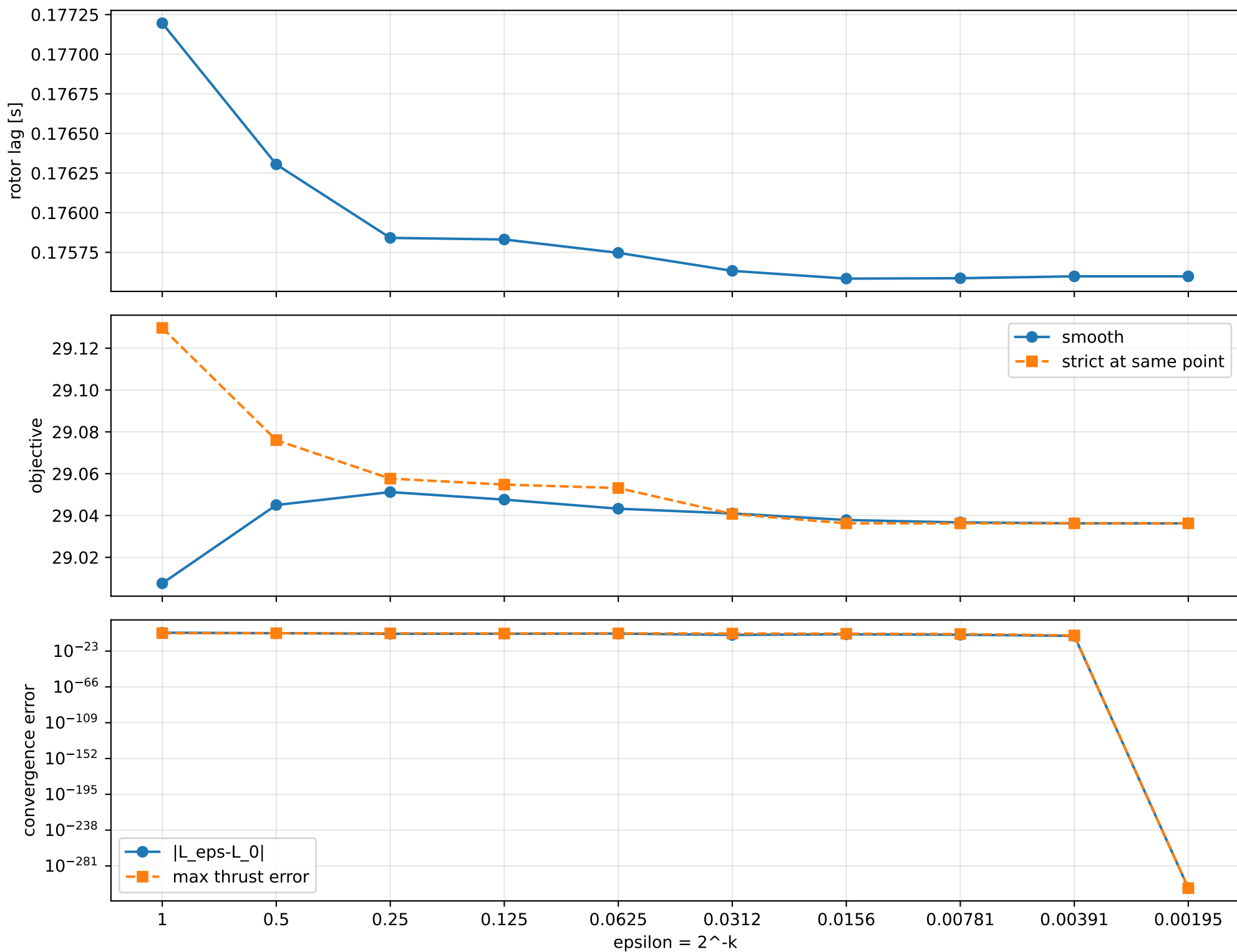
gyro [rad/s]



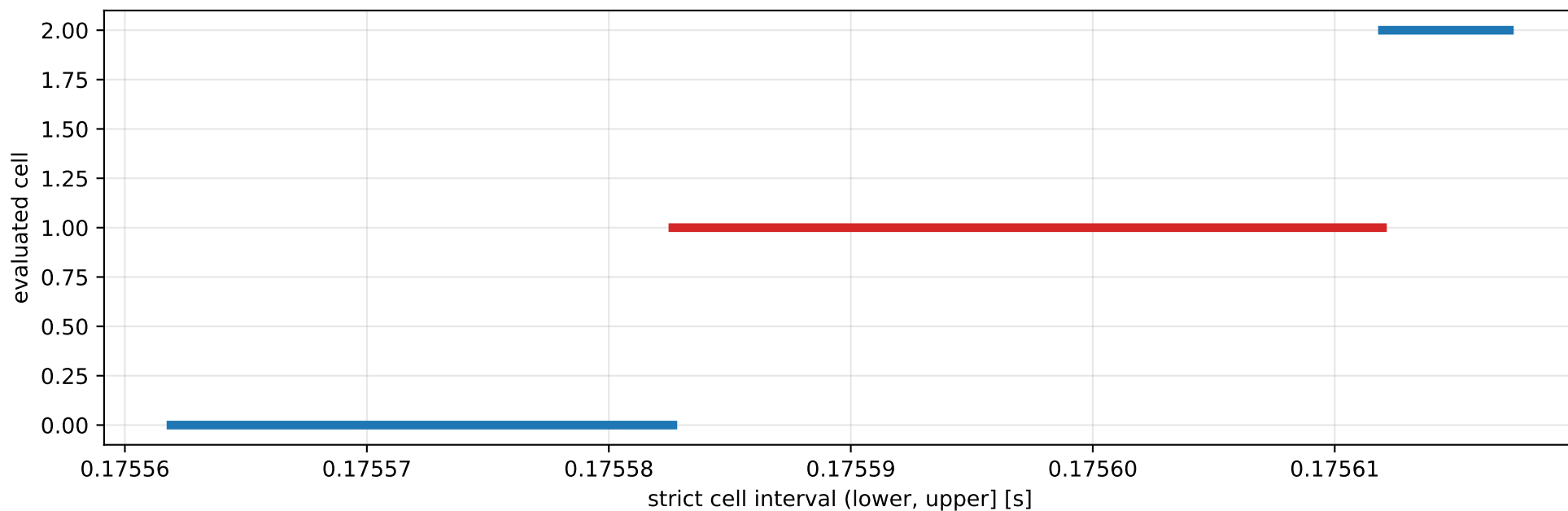
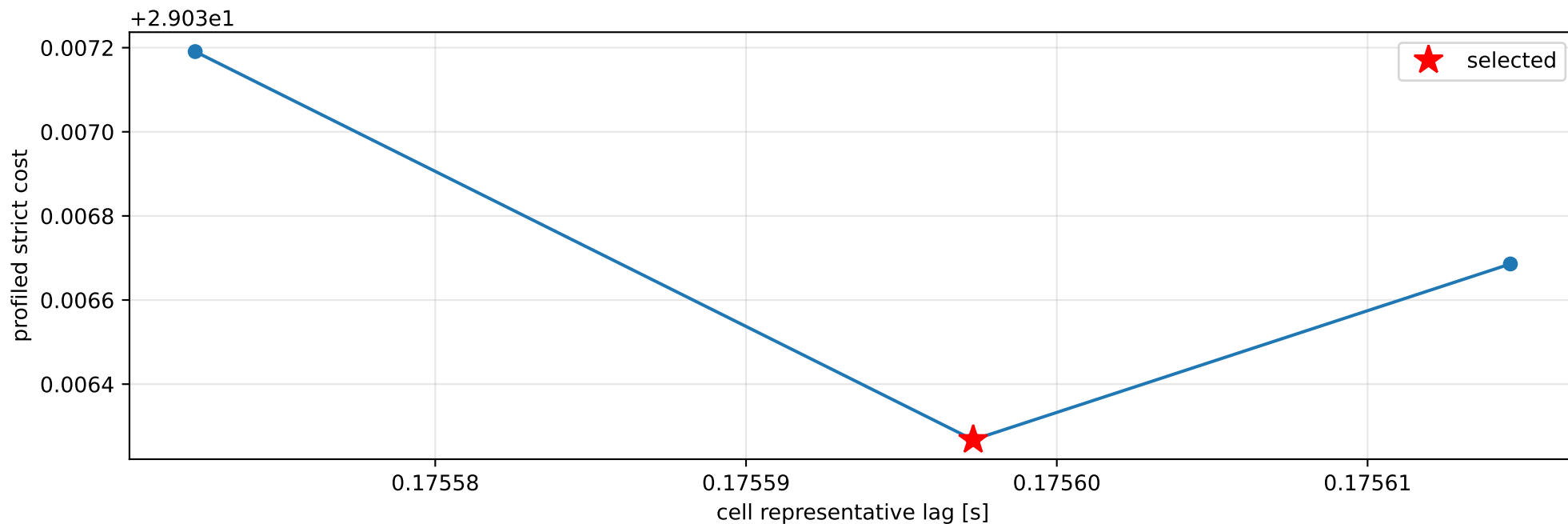
specific force [m/s²]



force_over_mass_rotor_1_nominal: smooth-to-strict continuation



force_over_mass_rotor_1_nominal: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.17559853 s; data boundary=False

force_over_mass_rotor_1_nominal: parameter estimate

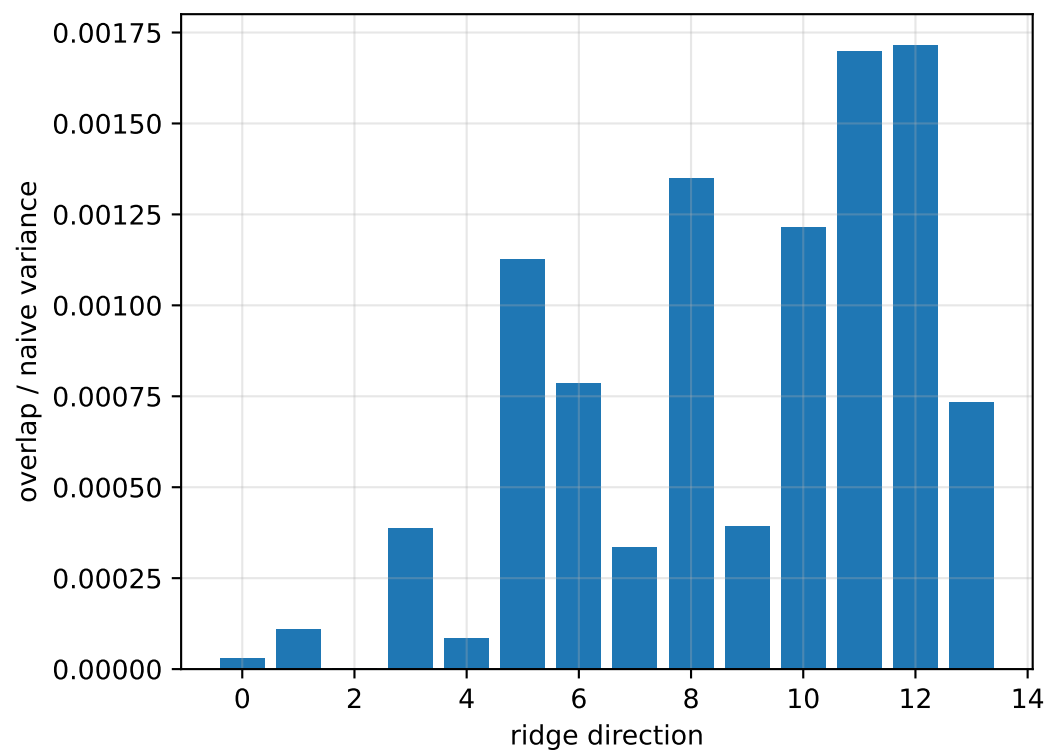
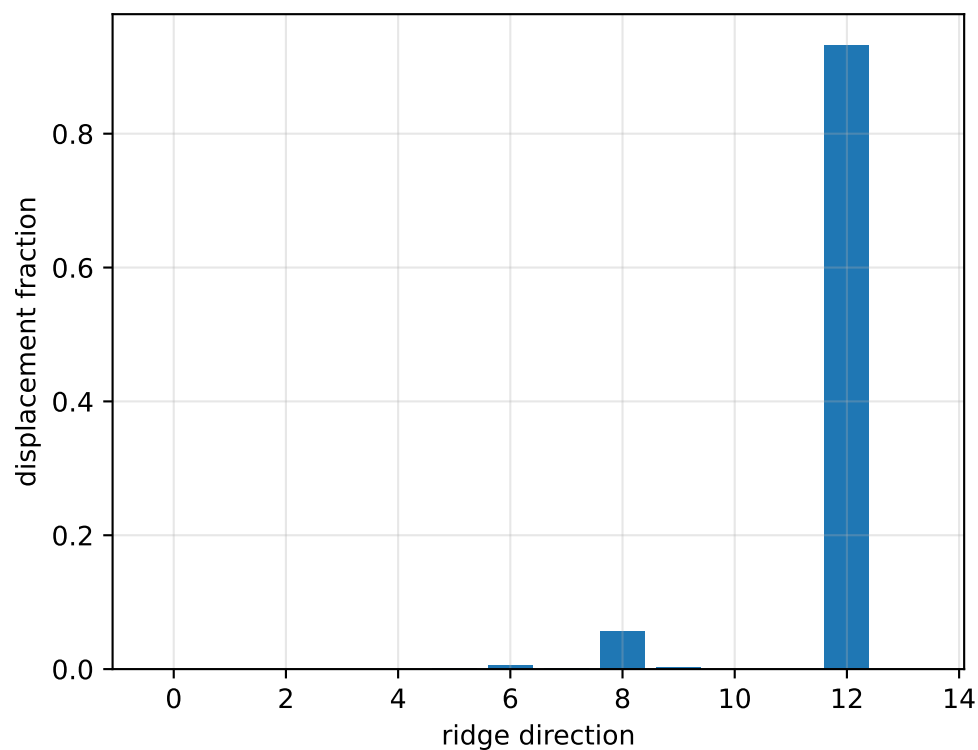
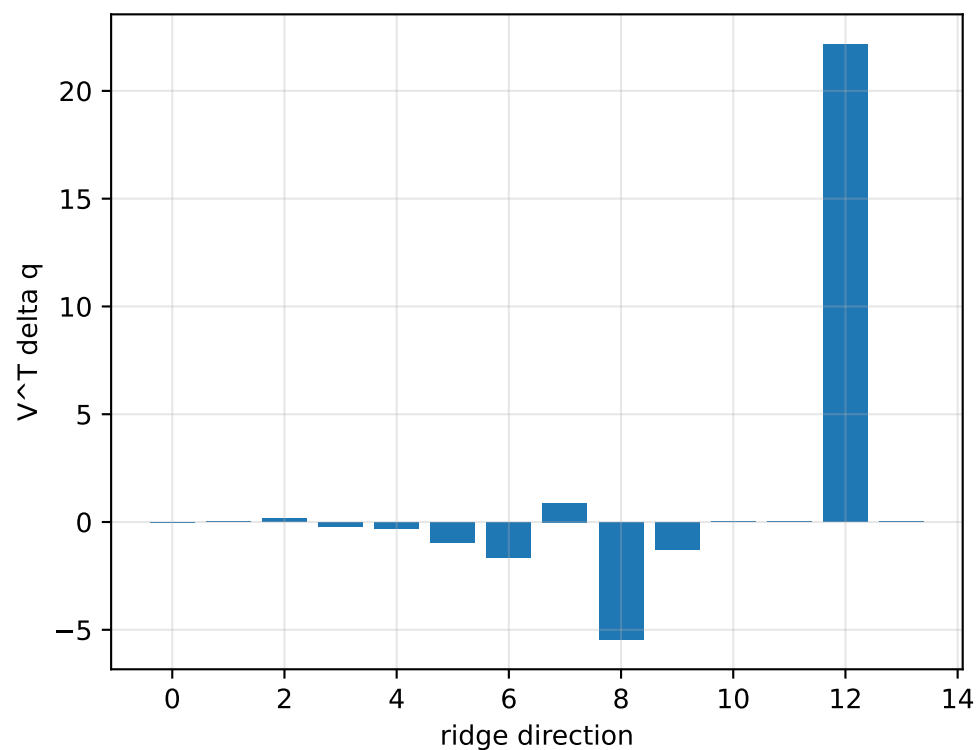
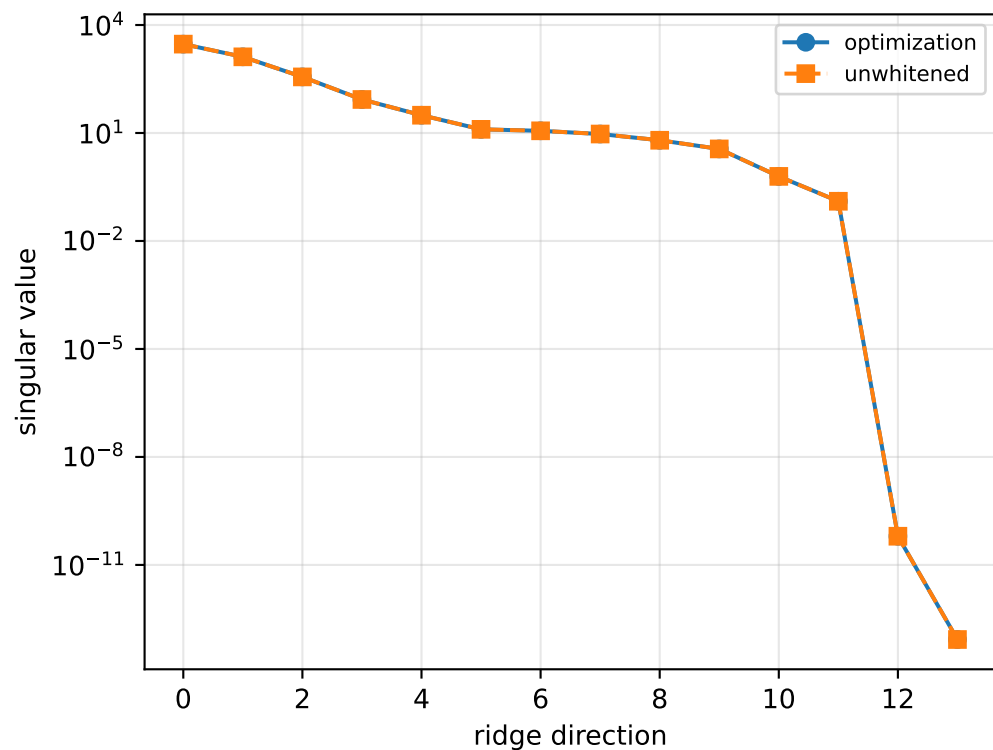
Case: force_over_mass_rotor_1_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 16.8382841
inertia [kg m²]:
[[4.20657443e+00 -3.95045729e-02 -3.59116858e-03]
 [-3.95045729e-02 1.86225623e+00 -1.02721748e-01]
 [-3.59116858e-03 -1.02721748e-01 2.34565295e+00]]
CoG body [m]: [-0.00906622 -0.05531705 -0.00467136]
force effectiveness: [7.16035959 7.68725294 4.08916762 4.47973632]

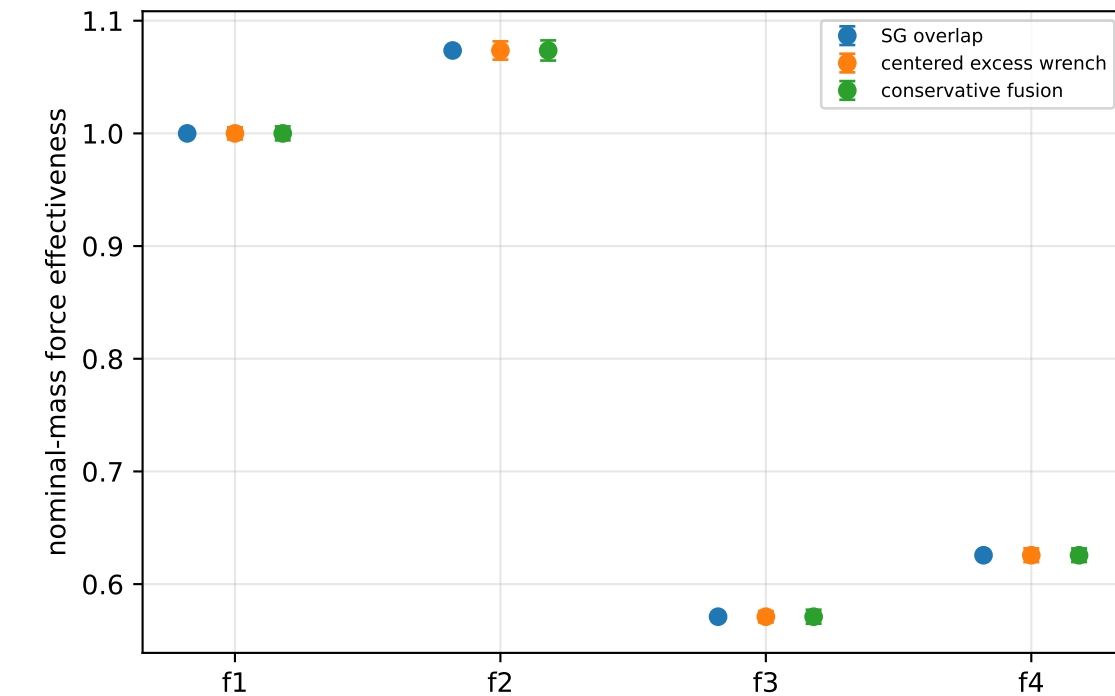
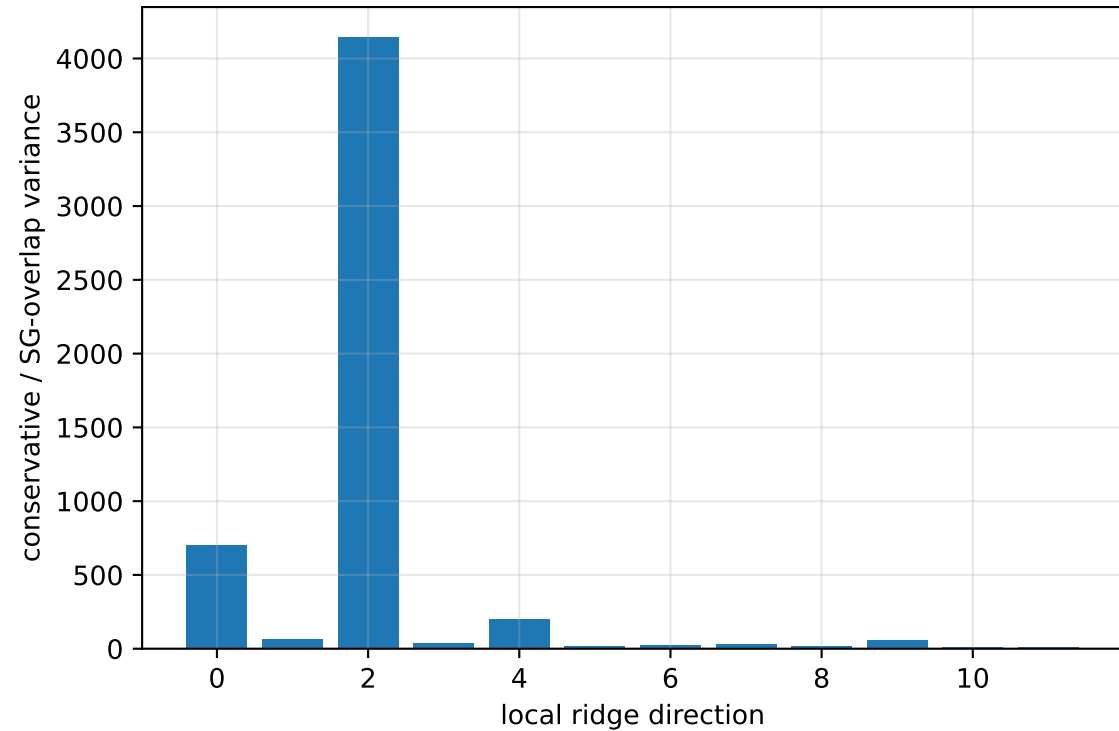
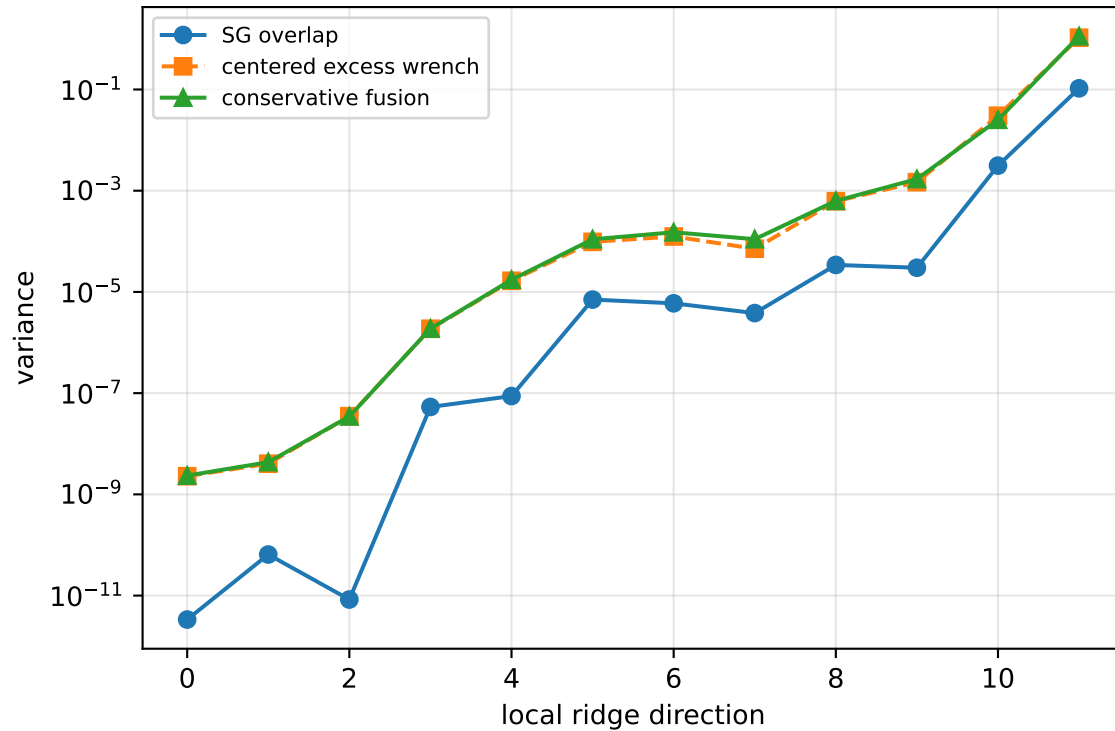
scale-free J/m [m²]:
[[2.49822037e-01 -2.34611630e-03 -2.13274023e-04]
 [-2.34611630e-03 1.10596556e-01 -6.10048790e-03]
 [-2.13274023e-04 -6.10048790e-03 1.39304750e-01]]
scale-free f/m: [0.42524283 0.45653422 0.24284943 0.2660447]

rotor lag cell: (0.175582647, 0.175611973] s
representative: 0.17559731 s

force_over_mass_rotor_1_nominal: ridge spectrum (rank 12, nullity 2, ||Jv||=7.64e-14)



force_over_mass_rotor_1_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 426.2$
wrench-to-acceleration recovery max error = $2.94\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

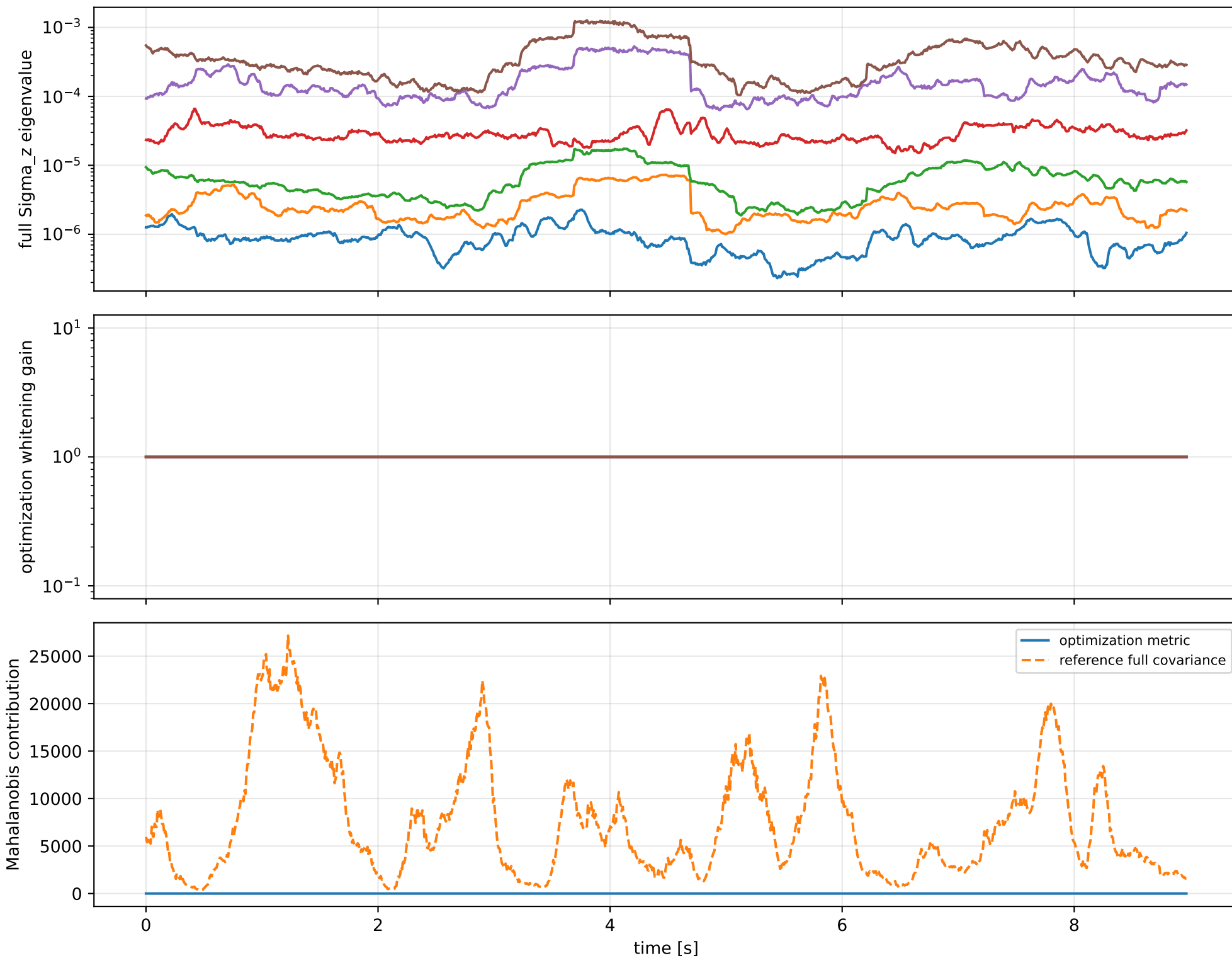
index 11: variance=1.128, ratio=10.69
 $\log_{\text{mass}} = +0.000156$; second_moment_diag_1=-0.0235
second_moment_diag_2=+0.0221; second_moment_diag_3=+0.00124
second_moment_offdiag_12=-0.993; second_moment_offdiag_13=+0.118
second_moment_offdiag_23=-0.00155; cog_x=+0.00042
cog_y=-0.000195; cog_z=+0.000315
log_force_eff_1=-0.00116; log_force_eff_2=+0.00284
log_force_eff_3=+0.000533; log_force_eff_4=-0.00215

index 10: variance=0.02517, ratio=8.041
 $\log_{\text{mass}} = +0.0013$; second_moment_diag_1=+0.00329
second_moment_diag_2=-0.0166; second_moment_diag_3=+0.00435
second_moment_offdiag_12=+0.117; second_moment_offdiag_13=+0.992
second_moment_offdiag_23=-0.0136; cog_x=+0.004
cog_y=-0.00067; cog_z=-0.000507
log_force_eff_1=-0.0107; log_force_eff_2=+0.0177
log_force_eff_3=+0.025; log_force_eff_4=-0.0245

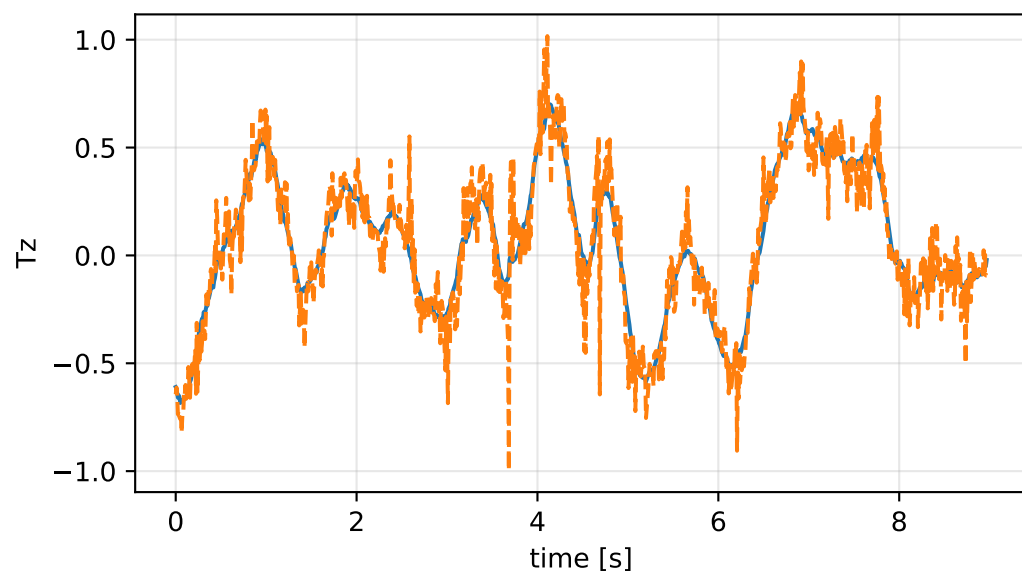
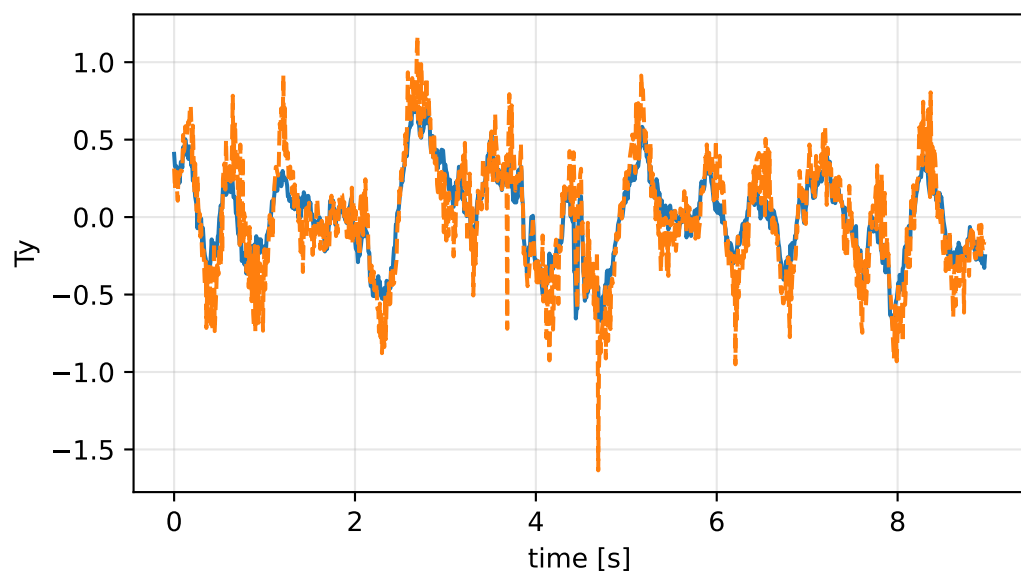
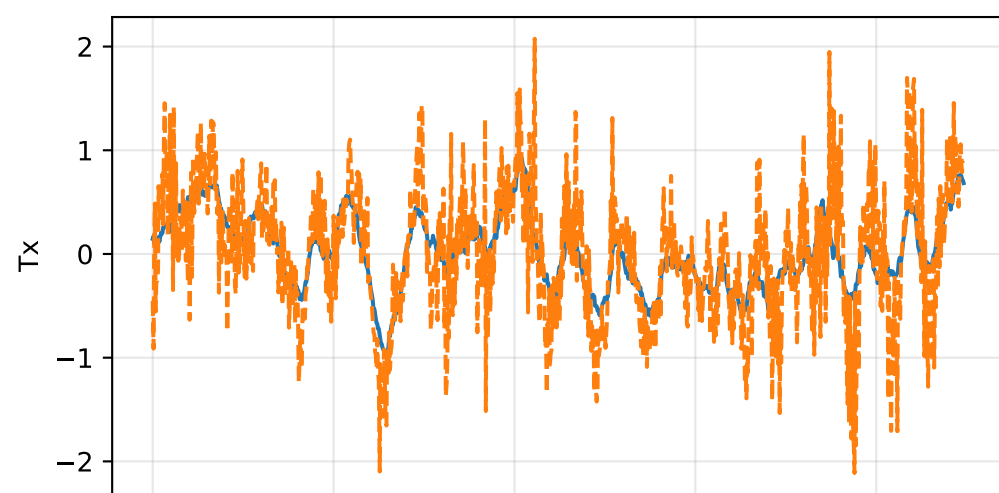
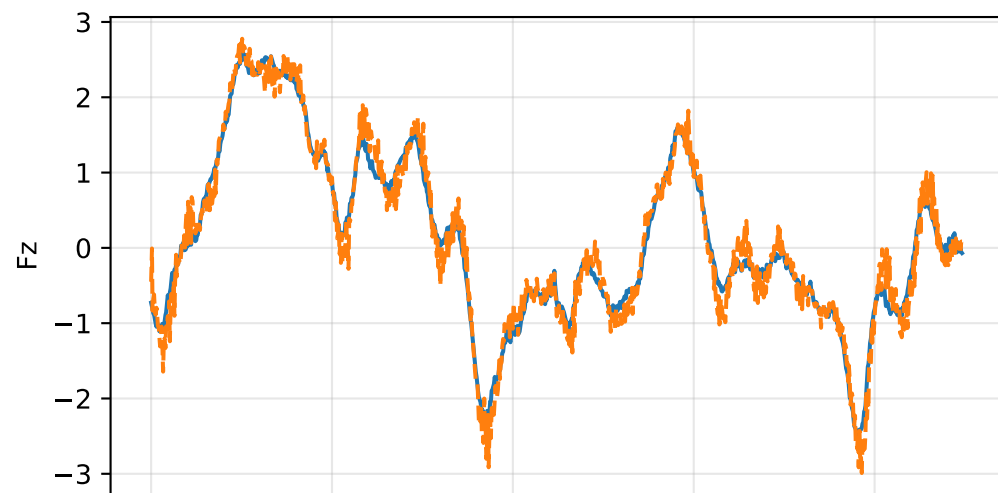
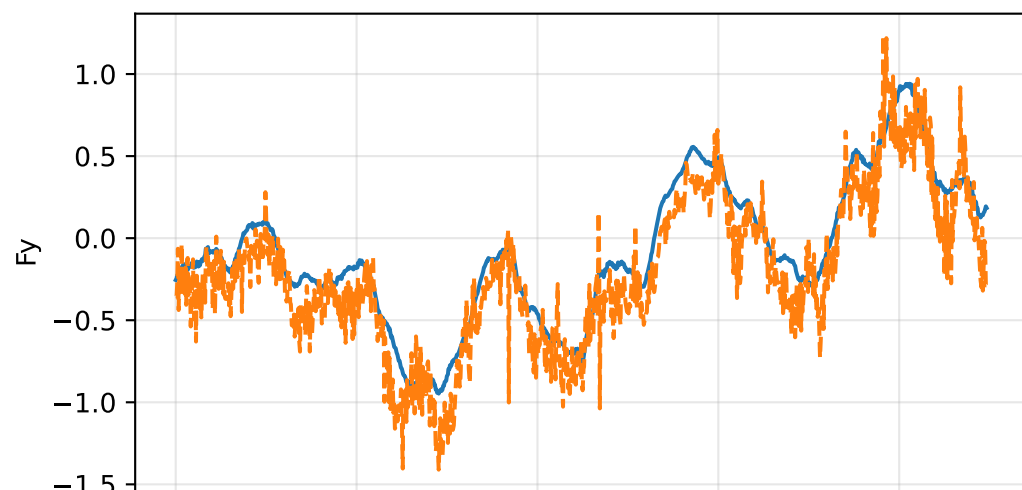
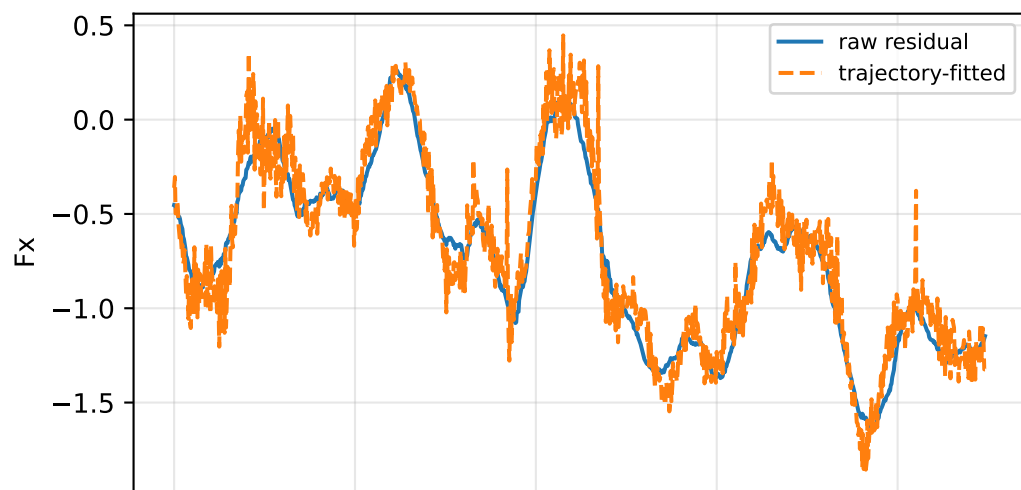
index 9: variance=0.001698, ratio=56.5
 $\log_{\text{mass}} = +0.171$; second_moment_diag_1=-0.000263
second_moment_diag_2=-0.898; second_moment_diag_3=+0.0539
second_moment_offdiag_12=-0.0215; second_moment_offdiag_13=-0.0157
second_moment_offdiag_23=-0.214; cog_x=-0.0042
cog_y=-0.00176; cog_z=+0.00922
log_force_eff_1=+0.204; log_force_eff_2=+0.151
log_force_eff_3=+0.145; log_force_eff_4=+0.173

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_rotor_1_nominal: optimization vs reference covariance



force_over_mass_rotor_1_nominal: wrench / closure (force max 4.95e-14, torque max 3.89e-16)



force_over_mass_rotor_1_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_rotor_1_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass

SHA256: 6d6d7476c7c2cc90b6dbb115ea73393bd5979488f12a4de1ccf810600f0d2db0

data objective: 29.0362674335

prior objective: 2.1959911698e-07

total objective: 29.0362676531

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_rotor_1_nominal (force_effectiveness_over_mass)

components: rotor_1

target: [0.42524282]

std: [1.e-05]

final: [0.42524283]

error: [6.62720329e-09]

standardized residual: [0.00066272]

factor objective: 2.1959911698e-07